KPS PUBLIC SCHOOLS • STUDENT TECH BENCH PROGRAM

Certification Guide

Tier 3 — Advanced Repair

T3

System boards, daughterboards, heat sinks, liquid damage and full transplants.

In this guide

1 About this certification and how to earn it

2 What you need to know (study modules)

3 Skills checklist

4 Supervised practice log

5 Knowledge check

6 Practical exam

7 Sign-off

Name: _____ Started: _____

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1 About this certification

What it lets you do

- Take Tier 3 tickets: system board and daughterboard replacement, heat sinks and fans, liquid damage, full top-cover transplants.
- Open the Senior-only guides in the portal.
- Mentor Tier 1 and Tier 2 students and check off their repairs.
- Decide whether a device is repairable or should go to the manufacturer (with the lead tech).

Before you start. Tier 2 — Challenging Repair, held for at least 4 weeks with no comebacks.

How to earn it



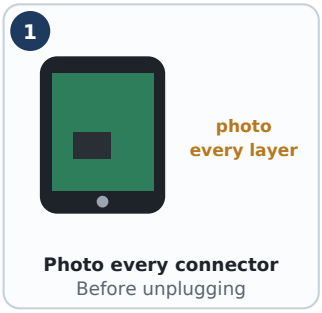
Step	What you do
Study	Modules 1–5 below, and the system board, heat sink and spill sections of the guides for your bench's main models.
Practise	Do 5 supervised Tier 3 repairs: at least 2 board or daughterboard swaps, 1 heat sink, and 1 liquid-damage response or full transplant.
Mentor	Coach a Tier 1 student through a full repair and check it off.
Knowledge check	Score 90% or better (14 of 15).
Practical exam	Pass the observed practical with no automatic-fail items.
Sign-off	Lead tech or manager signs and awards the certification in the portal.

Certification is earned, not given for time served. Every requirement below must be signed off by a lead tech or manager.

A certification can be paused by a manager after a safety violation or repeated comebacks (repairs that come back with the same fault).

When every box is signed, the manager awards the certification on the portal's Certifications page. That's what unlocks the matching tickets and guides.

2 What you need to know



Repairs in this tier, and where the guides are

Open the device in the portal's Learn tab and pick the topic. Each topic opens at the right section of that model's repair guide. Read the device's guide every time: screw counts and steps differ between models.

Learn tab topic	Covers
Motherboard Replacement	System board (Senior guide)
Daughterboard Replacement	I/O board (Senior guide)
Heatsink Replacement	Heat sink and fan (Senior guide)
(guide section) Liquid spill response	Liquid damage
Keyboard Replacement / Top Assembly on Level 3 models	Full palm rest transplant
Lenovo Duet EDU G2: screen, battery, camera	Glued tablet repairs (lead tech present)

Module 1 — What counts as Tier 3

- The district fee schedule puts these in **Tier 3 (Advanced Repair)**: motherboard, daughterboard, and other major parts as assessed.
- A mix of Tier 1 and Tier 2 items in one incident can also be billed at Tier 3. Tier 4 (device replacement) covers lost, stolen, beyond repair and not returned devices; that's a lead tech and office decision.

Module 2 — Is it really the board?

- Most "dead boards" are a battery, charger, cable or connector. Before ordering a board, the ticket must show: S1, S2, S3 attempted; known-good charger and battery; every connector reseated.
- USB-C ports, RAM and storage are soldered on Chromebooks. A dead USB-C port or failing storage means a system board (or warranty).
- Burn marks, melted plastic or a burnt smell: stop and escalate.

Module 3 — Board swaps

- Photograph every connector before unplugging it. Label tape helps.
- Move the parts that stay with the device: heat sink (with new thermal material), SSD if socketed, shields, tape.
- Lift boards by their edges, ESD strap on, onto the mat.
- Identity: the serial number and enrollment live on the board. The lead tech updates the asset record, handles deprovisioning in the Admin console, and checks whether the new board ships with the serial programmed. Some Lenovo boards need vital product data written at a depot.
- After the swap: recovery if needed, re-enroll, full diagnostics, every port and peripheral tested.

Module 4 — Heat and thermal paste

- Loosen heat sink screws in the order printed on the heat sink (numbers or letters). Tighten in the same order.
- Clean all old paste from the processor and heat sink with isopropyl alcohol and a lint-free cloth. Apply a pea-size dot of new paste, or the supplied pad.
- Never power on a board without its heat sink.

Module 5 — Liquid damage and transplants

- Liquid: force off, unplug, tent it keys-down, open and unplug the battery as soon as possible, dry 24–48 hours. Never power on to test. No rice, no hair dryer.
- Look for white or green corrosion. Clean small areas with isopropyl alcohol and a soft brush; heavy corrosion on the board means escalate.
- Full transplants (palm rest with built-in keyboard, glued tablets): plan the order from the guide, one mat section per part, test everything before closing.
- Glued tablets (Lenovo Duet): heat, suction cup, shallow pick away from the display cable, battery out first. Lead tech present.

3 Skills checklist

An evaluator initials each skill after watching you do it correctly, on your own, at least once. Practice repairs count.

#	Skill	Date	Evaluator
1	Documents a full "is it the board?" check before ordering		
2	Photographs and unplugs every connector on a system board		
3	Removes and refits a system board by the edges, ESD-safe		
4	Transfers the heat sink with new thermal paste in the correct order		
5	Replaces a daughterboard / I/O board and its ribbons		
6	Replaces a fan		
7	Runs recovery, re-enrollment and full diagnostics after a board swap		
8	Handles a liquid-damage device correctly from first response to dry-out		
9	Completes a full top-cover (palm rest) transplant		
10	Checks off another student's repair as a mentor		
11	Explains when to escalate to the manufacturer or lead tech		

4 Supervised practice log

Log every supervised practice repair (or front-desk transaction). The supervisor signs each line.

#	Date	Device / asset tag	Repair or task	Ticket #	Supervisor
1					
2					
3					
4					
5					
6					

5 Knowledge check

15 questions. Circle one answer for each. Closed book. The evaluator marks it with the key. Pass mark is on page 1.

1. Which is a Tier 3 repair?

A Battery
B Hinge
C Daughterboard
D Bezel

2. Before ordering a system board, the ticket must show…

A S1, S2, S3, known-good charger/battery, connectors reseated
B Nothing else
C A photo of the student
D Two weeks waiting

3. A USB-C port is loose and won't charge. What's the repair?

A Resolder the port at the bench
B Replace the battery
C Replace the I/O board
D System board or warranty

4. In what order do you loosen heat sink screws?

A Any order
B The order printed on the heat sink
C Clockwise from the fan
D All at once

5. How much new thermal paste?

A Fill the whole area thickly
B A pea-size dot, or the supplied pad
C None
D Reuse the old paste

6. A liquid-damaged device comes in. What should you never do?

A Unplug the battery
B Tent it to drain
C Dry it 24–48 hours
D Power it on to test

7. Where do the serial number and enrollment live?

A On the system board
B On the battery
C In the keyboard
D On the screen

8. After a board swap, what must happen before the device goes back?

A Nothing
B A new battery
C Recovery if needed, re-enroll, full diagnostics
D A new screen

9. How do you handle a system board?

A By the chips
B With metal pliers
C Any way, it's sturdy
D By its edges, ESD strap on

10. You see burn marks on the board. What do you do?

A Clean them off
B Stop and escalate
C Swap the battery
D Ignore it

11. Can you power on a board without its heat sink to test it?

A Yes, briefly
B Only on Intel
C No, never
D Only with a fan

12. A screen and a camera are on one ticket. Who decides the tier?

A The lead tech, using the fee schedule
B The student
C Whoever is fastest
D The parts vendor

13. During a glued-tablet screen removal, how deep does the pick go?

A Shallow, under about 3 mm, away from the display cable
B As deep as possible
C Through to the battery
D It doesn't matter

14. White or green crust on a board after a spill means...

A It's dust
B It's fine
C Corrosion: clean lightly or escalate
D Thermal paste

15. As a mentor, what do you do before checking off a student's repair?

A Trust them
B Check only the ticket
C Nothing
D Run the guide's verify checks and look inside before closing

Score	Pass mark	Result	Evaluator / date
____ / 15	90% or better (14 of 15)	Pass / Retake	

6 Practical exam

An evaluator watches you do each task. Time allowed: about 90 minutes. You may use the repair guides and your notes, but you work on your own.

Task	What you'll do	To pass	P / F
Board diagnosis	Given a "dead" device, decide whether the board is at fault.	Works through S1/S2/S3, charger, battery and connectors in order; correct conclusion; notes written.	
Board or daughterboard swap	Swap a board (or daughterboard) on a practice unit.	Photos, ESD-safe handling, every connector reseated, device boots.	
Heat sink	Remove and refit a heat sink with new paste.	Correct order, clean surfaces, correct amount, no power without the heat sink.	
Liquid response	Talk through and demonstrate the first 10 minutes of a spill.	Force off, unplug, tent, battery out, no power-on, correct dry time.	
Mentor check	Check off a planted student repair that has one hidden fault.	Finds the fault, explains it clearly, records the check-off correctly.	

Automatic fail. Any one of these ends the exam: Powering on a board without a heat sink; Powering on a wet device; Handling a board without ESD protection; Ordering a board without the documented checks; Prying near a battery or display cable on a glued device. You can retake it after a week of extra practice.

7 Sign-off

Requirement	Done	Signed
Study	☐	
Practise	☐	
Mentor	☐	
Knowledge check	☐	
Practical exam	☐	
Prerequisite: Tier 2 — Challenging Repair	☐	

Candidate	
Evaluator (lead tech or manager)	
Date awarded	
Awarded in the portal (Certifications page) by	

Managers: award it on the portal's Certifications page so it shows on the student's portfolio and unlocks the matching tickets and guides.